

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO3: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 30%

Dr G. A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Code of Conduct:

I K.Neeha(192425245) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:K.Neeha

1. Reinforcement Learning Framework – Agent–Environment Interaction (5 Marks)

Reinforcement Learning (RL) is a machine learning approach in which an **agent learns by interacting with an environment**. At every step, the agent observes the current state, selects an action, receives a reward, and moves to a new state.

Agent–Environment Interaction

Component	Description	Example – Robot Navigation
Agent	System that makes decisions	Robot
Environment	Surrounding world in which the agent operates	Grid/Map

Component	Description	Example – Robot Navigation
State (S)	Current situation of the agent	Robot's current position
Action (A)	Decision taken by the agent	Up, Down, Left, Right
Reward (R)	Feedback received after an action	+10 for reaching destination
Next State (S')	New situation after the action	New robot position
Policy (π)	Strategy used to select actions	Best direction for each position

RL Interaction Process

Step Process

- 1 Agent observes the current **state**
- 2 Agent selects an **action**
- 3 Environment executes the action
- 4 Agent receives a **reward**
- 5 Environment moves to the **next state**
- 6 Agent learns and improves its future decisions

Flow:

State → Action → Environment → Reward + Next State → Learning → Action

Importance of States, Actions and Rewards

Element	Importance
State	Helps the agent understand the current situation.
Action	Represents the possible decisions available to the agent.
Reward	Provides feedback about whether the selected action was good or bad.
Cumulative Reward	Helps the agent select actions that produce better long-term results.

Cumulative Reward

The RL agent does not focus only on the immediate reward. It attempts to maximize the **total reward received over multiple steps**.

$$G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \dots$$

where γ (**gamma**) is the discount factor.

For example, a robot may take a longer path with small negative rewards but eventually receive a large positive reward for reaching the destination. Therefore, it learns to choose the path with the **highest cumulative reward**.

2. Decision-Making Using Markov Decision Process (MDP)

A **Markov Decision Process (MDP)** is a mathematical framework used to model sequential decision-making problems where the result of an action depends on the current state and the selected action.

An MDP is represented as:

$$MDP = (S, A, P, R, \gamma)$$

Components of MDP

Component	Symbol	Description
State Space	S	Set of all possible states of the environment
Action Space	A	Set of actions available to the agent
Transition Probability	$P(s' s, a)$	Probability of reaching next state s' from state s after action a
Reward Function	$R(s, a, s')$	Reward received for taking action a and reaching next state
Discount Factor	γ	Determines the importance of future rewards

Example: Autonomous Robot Navigation

MDP Component	Example
States (S)	Different positions of the robot
Actions (A)	Up, Down, Left, Right
Transition Probability (P)	Probability that the robot reaches the intended position
Reward (R)	+10 destination, -10 obstacle, -1 normal movement
Discount Factor (γ)	0.9
Goal	Reach the destination with maximum cumulative reward

Role of Each Component

1. Action Space:

Defines all possible actions the agent can perform. For example, a robot may have four actions: **Up, Down, Left and Right**.

2. Transition Probability:

Defines how the environment changes after an action. For example, if the robot moves right, there may be a **90% probability** of reaching the intended cell and a **10% probability** of moving to another cell.

3. Reward Function:

Provides feedback to the agent. Good decisions receive positive rewards, while undesirable decisions receive negative rewards.

4. Discount Factor:

Controls how much importance is given to future rewards.

γ Value Behaviour

$\gamma = 0$ Considers only immediate rewards

γ close to 1 Gives high importance to future rewards

$\gamma = 1$ Future rewards are considered without discounting

MDP Decision-Making Flow

Stage	Description
1. Observe	Agent observes current state S
2. Select	Agent selects action A
3. Transition	Environment moves to next state S'
4. Reward	Agent receives reward R
5. Update	Agent updates its knowledge/policy
6. Repeat	Process continues until the goal is achieved

Conclusion

Thus, **RL uses agent–environment interaction to learn optimal decisions**, while an **MDP mathematically represents the states, actions, transition probabilities, rewards, and discount factor** involved in the decision-making process. The use of **cumulative reward** enables the agent to make decisions that are beneficial not only immediately but also in the long term.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism
Response to Questions	10	Responds accurately, promptly, and elaborates with relevant examples	Responds correctly but with limited depth	Partial responses; needs prompting	Unable to respond appropriately